

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Dhanush Kumar S	Reg. No.:	192421096
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Duplicate Record Detection System

A system compares each record with every other record using nested loops. Explain how nested loops impact time complexity. Determine the asymptotic time complexity of the algorithm. Analyze how execution time increases with larger datasets. Interpret efficiency using asymptotic notation.

Q2

Application: Cloud Data Backup System

A cloud backup system divides data into smaller chunks recursively: $T(n) = 2T(n/2) + n \log n$. Apply the Master Theorem and determine the time complexity. Identify the correct case and justify your answer. Discuss how the algorithm scales with increasing data size and interpret using asymptotic notation.

Code of Conduct

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Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
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Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

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Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
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Student Name:	Dhilipan M	Reg. No.:	192411038
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Hierarchical File Compression System

A file compression system processes data using the recurrence $T(n) = 2T(n/3) + n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how performance scales with increasing file size. Interpret efficiency using asymptotic notation.

Q2

Application: Natural Language Processing System

An NLP system processes text data using the recurrence $T(n) = T(n/2) + n$. Apply the substitution method to expand the recurrence. Derive the final time complexity and simplify it. Analyze how performance changes with increasing text corpus size. Interpret efficiency using asymptotic notation.

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Student Name:	D Chaitanya Kumar Reddy	Reg. No.:	192411094
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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Multi-Stage Authentication System

A security authentication algorithm reduces the problem size by half at each recursive stage and performs constant verification work at every level. Apply the substitution method to solve the recurrence $T(n) = T(n/2) + 1$. Show the detailed recursive expansion and derive the final asymptotic complexity. Analyze why the algorithm remains efficient over multiple levels of recursion. Interpret the result using asymptotic notation.

Q2

Application: IoT Data Processing System

An IoT system processes sensor data using the recurrence $T(n) = 3T(n/3) + n \log n$. Apply the Master Theorem and compare $f(n)$ with $n^a \log_b n$. Identify the correct case and justify the selection. Analyze how the system behaves for large IoT datasets. Interpret the efficiency using asymptotic notation.

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Student Name:	Dhorakonda Suresh	Reg. No.:	192525305
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Online Billing System

A billing system processes transactions using the recurrence $T(n) = T(n-1) + 1$. Apply the substitution method to expand the recurrence step-by-step. Derive the final time complexity and simplify it. Analyze how the system performs as the number of transactions increases. Interpret efficiency using asymptotic notation.

Q2

Application: Video Processing Pipeline

A video processing system follows the recurrence $T(n) = T(n/2) + \log n$. Apply the substitution method and expand the recurrence. Derive the final time complexity and simplify the expression. Analyze performance for high-resolution video processing. Interpret efficiency using asymptotic notation.

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Student Name:	Edrin J S	Reg. No.:	192524041
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Credit Card Validation Server

A banking server validates transactions using three nested loops to check account details, balance history, and audit logs. Recall how multiple nested loops affect complexity, explain the growth rate, and determine both the time complexity and space complexity for this system.

Q2

Application: Online Banking Fraud Detection

A fraud detection system analyzes transactions recursively: $T(n) = T(n/2) + 1$. Apply the Master Theorem and determine the time complexity. Identify the appropriate case and justify. Explain how this efficiency impacts real-time fraud detection performance.

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Gnana Priyan G	Reg. No.:	192521393
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Social Media Trend Analysis System

A social media system analyzes posts using the recurrence $T(n) = 2T(n/2) + n \log n$. Apply the Master Theorem to determine the time complexity. Identify parameters and analyze the space complexity.

Q2

Application: Smart Grid Energy Distribution System

A smart grid system processes energy distribution data using the recurrence $T(n) = 4T(n/2) + n \log n$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the case selection. Analyze how the system scales with increasing network size. Interpret the result using asymptotic notation.

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Student Name:	Hariharan B	Reg. No.:	192511394
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Search Engine Optimization System

A search engine processes queries using the recurrence $T(n) = T(n/2) + 1$. Apply the substitution method to expand the recurrence step-by-step. Derive the final time complexity and simplify the result. Analyze how the recursion depth influences the performance of the system. Interpret the result using asymptotic notation and explain why this approach is efficient for large datasets.

Q2

Application: Rank Comparison System

A university ranking system compares each student's performance with every other student to determine overall rankings using a pairwise comparison approach. Explain how nested comparisons influence the time complexity of the algorithm and describe the growth rate as the number of students increases. Determine the asymptotic time complexity using Big-O notation and justify your reasoning. Additionally, analyze the space complexity of the system, considering storage of intermediate comparisons. Interpret how this approach impacts performance for large datasets and discuss its efficiency.

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Student Name:	Jabastian R	Reg. No.:	192472138
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Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Genome Sequencing Analysis

A genome sequencing algorithm is modeled as: $T(n) = 3T(n/2) + n$. Apply the Master Theorem to determine time complexity. Clearly identify parameters and justify the case. Analyze how scalability affects large biological datasets.

Q2

Application: Cloud Log Processing System

A cloud-based system processes large volumes of log data using the recurrence $T(n) = 3T(n/2) + n$. Apply the Master Theorem to determine the time complexity of the algorithm. Clearly identify the values of a , b , and $f(n)$, and justify the selected case. Analyze how the algorithm performs when log data grows exponentially in distributed environments. Interpret the final result using asymptotic notation and explain its impact on system efficiency.

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Student Name:	Jayaprakash Nivas	Reg. No.:	192424238
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Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Insurance Verification System

An insurance verification system uses multiple nested loops to validate policy details, customer history, and claim records simultaneously. Explain how nested loops contribute to higher time complexity and describe the resulting growth rate. Determine the asymptotic time complexity using Big-O notation and justify your reasoning clearly. Analyze the space complexity, including memory used for storing intermediate validation results. Interpret how this approach performs for large datasets and discuss its efficiency limitations.

Q2

Application: Online Transaction Processing System

A transaction system processes records using the recurrence $T(n) = T(n-1) + \log n$. Solve the recurrence using substitution method and derive the time complexity. Also determine the space complexity and explain growth behavior.

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Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
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Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Kampasati Venkatesh	Reg. No.:	192425102
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Insurance Claim Verification System

An insurance system uses three nested loops to verify policy details, claim history, and supporting records. Recall how triple nested loops affect complexity, explain the rate of growth, and determine both the time complexity and space complexity for this process.

Q2

Application: Network Access Control System

A network access system checks each user request sequentially against an access control list. Define linear search, explain how the algorithm works, and determine both the time complexity and space complexity as the number of requests increases.

Code of Conduct

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Signature of the Student: _____

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Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Keerthipati Mithil Varma	Reg. No.:	192425142
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
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- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Network Routing Optimization System

A routing system processes nodes using the recurrence $T(n) = 3T(n/2) + n$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and analyze the space complexity.

Q2

Application: Distributed Data Processing System

A distributed system processes data blocks using the recurrence $T(n) = T(n/2) + n$. Recall recurrence solving methods, explain how work grows across recursive calls, and determine the asymptotic time complexity. Also analyze the space complexity of the algorithm.

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Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

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Date: _____	Date: _____	Date: _____

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Konakalla Naga Venkata Durga Prem	Reg. No.:	192411229
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
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- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Blockchain Transaction Verification System

A blockchain system verifies transactions using the recurrence $T(n) = 5T(n/5) + \log n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how performance changes as transaction volume increases. Interpret efficiency using asymptotic notation.

Q2

Application: Software Log Sorting System

One log sorting algorithm takes $O(n \log n)$ time, while another takes $O(n^2)$ time. Compare both algorithms using asymptotic notation, explain differences in growth rates, and decide which is better for large datasets. Also compare their space complexities and identify the more memory-efficient choice.

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Q1 (50 Marks)						
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Overall Total (100)						

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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
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Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Kunchala Venkatesh	Reg. No.:	192424326
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Social Media Feed Processing

A social media platform processes posts using a recursive algorithm $T(n) = T(n-1) + n$. Apply the substitution method to solve the recurrence. Expand step-by-step and derive the final complexity. Analyze how performance degrades with large inputs. Interpret the result using asymptotic notation.

Q2

Application: Smart Warehouse Inventory System

A warehouse system processes inventory using the recurrence $T(n) = 4T(n/2) + n \log n$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and analyze the space complexity.

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Q1 (50 Marks)						
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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Kunuthuru Deepthi	Reg. No.:	192472211
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
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- Justify each step in recurrence solving using appropriate methods.
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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Image Segmentation System

An image segmentation system processes data using the recurrence $T(n) = 2T(n/2) + n$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the case selection. Analyze how performance changes with increasing image size. Interpret efficiency using asymptotic notation.

Q2

Application: Smart Library Book Retrieval System

A library system searches for books using the recurrence $T(n) = T(n/2) + \log n$. Solve the recurrence using substitution method and determine the time complexity. Also analyze the space complexity.

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Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Karnesh V K	Reg. No.:	192524296
Section / Batch:		Date:	

Instructions to Students

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Hospital Patient Record System

A hospital system compares every patient record with every other record to detect duplicate entries using nested loops. Recall what nested loops imply in complexity analysis, explain why this approach increases execution time, and determine both the time complexity and space complexity using asymptotic notation.

Q2

Application: Dictionary Search System

A digital dictionary system uses divide-and-conquer to search entries. Explain how input size reduces at each step. Determine the time complexity of the algorithm. Analyze its efficiency for large datasets. Interpret the result using asymptotic notation.

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Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Kruthika Sre S	Reg. No.:	192525360
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Product Sorting System

A system compares sorting algorithms with complexities $O(n \log n)$ and $O(n^2)$. Analyze both using asymptotic notation and compare their growth rates. Explain which algorithm is more efficient for large datasets. Justify your answer based on scalability. Interpret efficiency clearly.

Q2

Application: Cloud Storage Processing System

A cloud storage system processes data using the recurrence $T(n) = T(n/2) + n$. Apply the substitution method to expand the recurrence. Derive the final time complexity and simplify it. Analyze how performance scales with increasing data size. Interpret efficiency using asymptotic notation.

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Q1 (50 Marks)						
Q2 (50 Marks)						
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Signature: _____	Signature: _____	Signature: _____
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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
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ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Lohitha R	Reg. No.:	192524250
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Log Synchronization System

A log synchronization system processes distributed logs using the recurrence $T(n) = 2T(n/2) + \log n$. Apply the substitution method and expand the recurrence. Derive the final time complexity and simplify it. Analyze performance for large-scale systems. Interpret the efficiency using asymptotic notation.

Q2

Application: Banking Authentication System

A banking authentication system verifies users using the recurrence $T(n) = T(n/2) + 1$. Apply the substitution method and expand the recurrence. Derive the final time complexity. Analyze how recursion depth affects system performance. Interpret the result using asymptotic notation.

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Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

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Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
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ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	M Bharath Ram	Reg. No.:	192521481
Section / Batch:		Date:	

Instructions to Students

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Parallel Task Scheduling System

A distributed computing system schedules tasks using the recurrence $T(n) = 7T(n/2) + n$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how this affects system performance as the number of tasks increases. Interpret efficiency using asymptotic notation.

Q2

Application: Digital Dictionary Search Engine

A dictionary search system reduces the remaining search entries by half after each comparison. Define divide-and-conquer strategy, explain how input size reduces, and calculate the number of steps. Determine both the time complexity and space complexity of this search process.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
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Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Pavithra K	Reg. No.:	192411027
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Autonomous Vehicle Sensor System

A vehicle system processes sensor data using the recurrence $T(n) = 5T(n/2) + n^2$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and analyze the space complexity.

Q2

Application: Smart Traffic Signal Optimization System

A smart traffic management system processes vehicle flow data using a divide-and-conquer algorithm represented by the recurrence $T(n) = 2T(n/2) + n$. Apply the Master Theorem to determine the time complexity of the algorithm. Clearly identify the parameters a , b , and $f(n)$, and justify the appropriate case selection. Analyze how the performance of the system changes as the number of vehicles increases significantly in urban areas. Interpret the final result using asymptotic notation and explain how it affects real-time traffic signal optimization.

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Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Poli Karthik	Reg. No.:	192311198
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
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- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Online Library Search System

An online library processes search queries using the recurrence $T(n) = T(n-1) + \log n$. Apply the substitution method to expand the recurrence step-by-step and derive the final time complexity. Clearly show intermediate steps and simplify the expression. Analyze how the algorithm behaves when the number of documents increases significantly. Interpret the efficiency using asymptotic notation and discuss scalability.

Q2

Application: Intelligent Search Optimization System

A recursive search optimization algorithm reduces the input size by half and performs logarithmic work at each stage because of indexed search operations. Apply the substitution method to solve the recurrence $T(n) = T(n/2) + \log n$. Show all intermediate expansions and derive the final asymptotic time complexity. Analyze how the logarithmic work affects efficiency for large inputs. Interpret the result using asymptotic notation.

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Papagari Pushpa Charitha	Reg. No.:	192525371
Section / Batch:		Date:	

Instructions to Students

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Automated Document Processing System

A document processor handles one page at a time using the recurrence $T(n) = T(n-1) + 1$. Recall the meaning of a recurrence relation, explain how it expands step-by-step, and solve it using substitution method to determine the final time complexity. Also determine the space complexity of the recursive process.

Q2

Application: Satellite Data Transmission System

A satellite system divides transmitted data using the recurrence $T(n) = 2T(n/2) + n$. Apply the Master Theorem to determine the time complexity. Identify parameters and also determine the space complexity of the recursion.

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Q2 (50 Marks)						
Overall Total (100)						

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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Pooja Sri S	Reg. No.:	192571023
Section / Batch:		Date:	

Instructions to Students

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- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Cloud Backup Analysis System

A cloud system processes backup data using the recurrence $T(n) = T(n/2) + n$. Recall recurrence solving techniques, explain how the function grows, and determine the asymptotic time complexity using an appropriate method. Also analyze the space complexity of the recursive calls.

Q2

Application: Library Search System

A digital library system uses a divide-and-conquer strategy to locate books by repeatedly halving the search space. Explain how this reduction in input size affects the number of operations required. Determine the time complexity of the algorithm using asymptotic notation and justify your reasoning. Also analyze the space complexity, particularly in recursive implementations. Interpret how this approach improves efficiency compared to linear search for large-scale library systems.

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Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Pragadeshwar S P	Reg. No.:	192571061
Section / Batch:		Date:	

Instructions to Students

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Smart Classroom Attendance System

A smart attendance system checks each student ID exactly once while marking presence. Identify the concepts of time complexity and space complexity, explain how the number of operations and memory usage grow with input size, and determine the asymptotic complexities of this process.

Q2

Application: Remote Sensing Image Processor

A satellite image is divided into two equal parts and processed recursively, with additional linear work at each step: $T(n) = 2T(n/2) + n$. State the Master Theorem $T(n) = aT(n/b) + f(n)$, explain the role of parameters a , b , and $f(n)$, and apply it to determine the time complexity. Also analyze the space complexity of this recursive image processing method.

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Student Name:	Prasanna A	Reg. No.:	192525130
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: File Search System

A file search system uses a divide-and-conquer approach by reducing the search space by half in each step. Define logarithmic time complexity and explain how halving affects performance. Determine the time complexity of the algorithm. Analyze its efficiency for large datasets. Interpret the result using asymptotic notation.

Q2

Application: Autonomous Driving Data Processing System

An autonomous driving system processes sensor data using the recurrence $T(n) = 2T(n/2) + n^2$. Apply the Master Theorem to determine the time complexity of the algorithm. Clearly identify the parameters a , b , and $f(n)$, and justify the appropriate case selection. Analyze how the performance of the system is affected as the volume of sensor data increases. Interpret the result using asymptotic notation and explain its impact on real-time decision-making.

Code of Conduct

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Department of Computer Science and Engineering

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Student Name:	R Loknath	Reg. No.:	192425007
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: E-Library Search System

An e-library reduces the collection size by half after each comparison while searching for a title. Define divide-and-conquer strategy, explain how repeated halving reduces the problem size, and calculate the number of steps. Determine both the time complexity and space complexity of this search system.

Q2

Application: Image Processing System

An image processing algorithm divides images recursively: $T(n) = 4T(n/2) + n$. Apply the Master Theorem to find the time complexity. Clearly identify parameters and case selection. Analyze how performance changes with image size.

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Student Name:	Ramaihgari Puneeth	Reg. No.:	192372362
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Sensor Network Analysis System

A sensor network processes data using the recurrence $T(n) = 3T(n/3) + 1$. Apply the substitution method and expand the recurrence step-by-step. Derive the final time complexity and simplify it. Analyze how the system scales with increasing number of sensors. Interpret the result using asymptotic notation.

Q2

Application: Financial Analytics System

A financial system uses an algorithm with recurrence $T(n) = 2T(n/2) + n \log n$. Apply the Master Theorem and determine the complexity. Compare $f(n)$ with $n^{\log_b a}$ and justify the correct case. Analyze performance for large datasets. Interpret efficiency clearly.

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Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Shaik Thaheer	Reg. No.:	192425347
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Recursive Financial Summation Engine

A financial computation system reduces the input size by half and performs linear work at each recursive call to compute cumulative summaries. Apply the substitution method to solve the recurrence $T(n)=T(n/2)+n$. Expand the recurrence step-by-step and verify the final result. Analyze how the algorithm behaves for large-scale transaction data. Interpret the efficiency using asymptotic notation.

Q2

Application: Natural Language Processing Engine

An NLP engine processes text data using the recurrence $T(n)=T(n/2)+n$. Apply the substitution method to solve the recurrence. Expand the relation step-by-step and derive the complexity. Analyze scalability as text corpus size increases. Interpret the result using asymptotic notation.

Code of Conduct

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Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
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Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
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Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Sheik Mohammed Afnan A	Reg. No.:	192424278
Section / Batch:		Date:	

Instructions to Students

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Video Streaming Quality Adjustment System

A streaming system processes frames using the recurrence $T(n) = 2T(n/2) + 1$. Apply the Master Theorem to determine the time complexity. Also analyze the space complexity.

Q2

Application: Medical Diagnosis Support System

A medical diagnosis system analyzes patient data using the recurrence $T(n) = 2T(n/4) + n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how this affects performance for large hospital datasets. Interpret efficiency using asymptotic notation.

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Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

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Department of Computer Science and Engineering

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Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Samuvel P	Reg. No.:	192511280
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: E-Commerce Recommendation System

An e-commerce system uses two algorithms with complexities $O(n \log n)$ and $O(n^2)$. Analyze both using asymptotic notation. Compare their performance for large datasets. Explain growth rate differences and justify which is more efficient. Provide a conclusion based on scalability.

Q2

Application: Satellite Image Analysis System

A satellite image analysis system processes image regions using the recurrence $T(n) = 7T(n/2) + n^2$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how processing time grows with image resolution. Interpret efficiency using asymptotic notation.

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Santhosh G	Reg. No.:	192521035
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Multi-Source Log Processing System

A distributed log processing platform divides input into two equal parts and performs logarithmic work during the merging phase due to search and synchronization operations. Apply the substitution method to solve the recurrence $T(n) = 2T(n/2) + \log_2 n$. Expand the recurrence in detail and derive the final complexity. Analyze performance for large log volumes. Interpret the result using asymptotic notation.

Q2

Application: Cloud Storage Processing System

A cloud system processes files using $T(n) = 3T(n/2) + n$. Apply the Master Theorem to find the time complexity. Identify parameters and justify the case selection. Analyze how this impacts performance for large storage systems. Interpret efficiency using asymptotic notation.

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ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Tottireddy Gayathri	Reg. No.:	192425239
Section / Batch:		Date:	

Instructions to Students

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Digital File Search System

A file search tool repeatedly reduces the search space by half while locating a file in a sorted directory. Define logarithmic growth, explain how halving affects the number of steps, and determine both the time complexity and space complexity of this algorithm.

Q2

Application: Smart Agriculture Monitoring System

A smart agriculture system processes field sensor data using the recurrence $T(n) = 5T(n/2) + n^2$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how performance changes as the number of sensors increases. Interpret efficiency using asymptotic notation.

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ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Vankivolu. Vignesh Reddy	Reg. No.:	192425077
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Online Examination Evaluation System

An online examination system evaluates student responses using the recurrence $T(n) = 3T(n/3) + n^2$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how performance changes as the number of answer scripts increases. Interpret efficiency using asymptotic notation.

Q2

Application: AI Model Training Pipeline

An AI training system processes datasets using the recurrence $T(n) = 9T(n/3) + n^2$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how training time grows with dataset size. Interpret efficiency using asymptotic notation.

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Q1 (50 Marks)						
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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	V Mohammad Hussain	Reg. No.:	192524179
Section / Batch:		Date:	

Instructions to Students

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Network Security Monitoring System

A network security monitoring system scans logs sequentially to detect anomalies and potential threats in real time. Define linear time complexity and explain how the number of operations grows with the size of the log data. Determine the asymptotic time complexity and justify your reasoning. Analyze the space complexity, considering storage of logs and temporary variables. Interpret how this approach performs for large-scale systems and discuss its efficiency in continuous monitoring environments.

Q2

Application: Digital Image Filtering System

An image processing system follows the recurrence $T(n) = 4T(n/2) + n^2$. Apply the Master Theorem to determine the time complexity. Identify parameters and determine the space complexity.

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Q1 (50 Marks)						
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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Arvind Krishna N	Reg. No.:	192421061
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Digital Map Navigation System

A navigation system processes routes using the recurrence $T(n) = T(n/2) + 1$. Solve the recurrence using substitution method to determine the time complexity. Also analyze the space complexity of the recursive calls.

Q2

Application: Credit Card Validation System

A credit card validation system uses three nested loops to verify transactions. Explain how multiple nested loops affect time complexity. Determine the asymptotic complexity of the algorithm. Analyze how performance degrades for large datasets. Interpret efficiency using asymptotic notation. ####

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Student Name:	Bestha Akshaya	Reg. No.:	192424336
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Multimedia Rendering Engine

A multimedia rendering engine evaluates three algorithms with complexities $O(n)$, $O(n \log n)$, and $O(n^2)$. Analyze each algorithm using asymptotic notation and compare their growth rates. Explain how performance differs when processing high-resolution media files. Identify the most efficient algorithm for large-scale rendering tasks. Justify your answer based on computational efficiency and scalability. ###

Q2

Application: Social Media Feed Ranking

A social media platform ranks posts using: $T(n) = 4T(n/2) + n^2$. Apply the Master Theorem to compute the time complexity. Clearly specify parameters and case selection. Analyze scalability issues as the number of posts increases and interpret efficiency.

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Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Venkata Udaykumar	Reg. No.:	192411203
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Online Learning Recommendation System

An online learning platform processes user interaction data using the recurrence $T(n) = T(n-1) + n$. Apply the substitution method to expand the recurrence step-by-step and derive the final time complexity. Clearly show intermediate steps and simplify the expression. Analyze how the algorithm performs when the number of users grows rapidly. Interpret the efficiency using asymptotic notation and discuss whether the system is scalable for large educational platforms. ###

Q2

Application: Weather Forecasting Simulation System

A weather simulation system uses two algorithms with complexities $O(n \log^2 n)$ and $O(n^2 \log n)$. Analyze both using asymptotic notation. Compare their performance for large-scale simulations. Explain growth rate differences and justify which algorithm is more efficient. Provide a conclusion based on scalability.

Code of Conduct

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Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Bille Vinay Kumar	Reg. No.:	192373042
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
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- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Big Data Analytics Pipeline

A data analytics pipeline processes input using $T(n) = T(n-1) + \log n$. Apply the substitution method to solve the recurrence. Expand step-by-step and derive complexity. Analyze how the algorithm scales. Interpret efficiency using asymptotic notation.

Q2

Application: Search Engine Indexing

A search engine indexing algorithm is defined as: $T(n) = 2T(n/2) + n^2$. Apply the Master Theorem and determine the time complexity. Clearly explain the chosen case. Analyze how indexing time grows with web pages.

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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Billuritheerthi Reddy	Reg. No.:	192473026
Section / Batch:		Date:	

Instructions to Students

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Document Processing System

A document processing system processes files sequentially using the recurrence $T(n) = T(n-1) + 1$, where each step handles one document. Apply the substitution method to expand the recurrence relation step-by-step and derive the final time complexity. Clearly explain how the recurrence unfolds and simplifies into a closed-form expression. Analyze the space complexity, particularly in recursive implementations where function calls are stacked. Interpret how this linear approach affects scalability when processing a large number of documents.

Q2

Application: Password Lookup System

A secure password lookup system uses a binary search technique to locate user credentials in a sorted database by repeatedly dividing the search space into halves. Define logarithmic time complexity and explain how reducing the input size affects the number of operations performed. Determine the time complexity of the algorithm and justify it using asymptotic notation. Also analyze the space complexity, especially in recursive implementations of binary search. Finally, interpret how this method ensures efficient performance even for very large databases.

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CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Bogu Mallika	Reg. No.:	192324234
Section / Batch:		Date:	

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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Robotics Path Planning System

A robotic navigation system processes paths using the recurrence $T(n) = 2T(n/4) + n^2$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze performance for complex environments. Interpret efficiency using asymptotic notation.

Q2

Application: Weather Forecasting Simulation System

A weather simulation system uses two algorithms with complexities $O(n \log n)$ and $O(n^2 \log n)$. Analyze both algorithms using asymptotic notation and compare their growth rates. Explain how each algorithm performs as the dataset size increases. Determine which algorithm is more efficient for large-scale simulations. Justify your answer based on computational efficiency and scalability.

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Student Name:	Cherukuri Rahul Pramodh	Reg. No.:	192373039
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Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1

Application: Fraud Detection in Financial Systems

A fraud detection system processes transactions using the recurrence $T(n) = T(n-1) + n^2$. Solve the recurrence using substitution method and determine the time complexity. Also analyze the space complexity.

Q2

Application: Digital Payment Processing System

A digital payment system uses three different algorithms with complexities $O(n)$, $O(n \log n)$, and $O(n^2)$ to process transactions. Analyze each algorithm using asymptotic notation and compare their growth rates. Explain how each algorithm behaves as the number of transactions increases. Identify the most efficient algorithm for large-scale payment processing systems. Justify your answer by considering both performance and scalability requirements.

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